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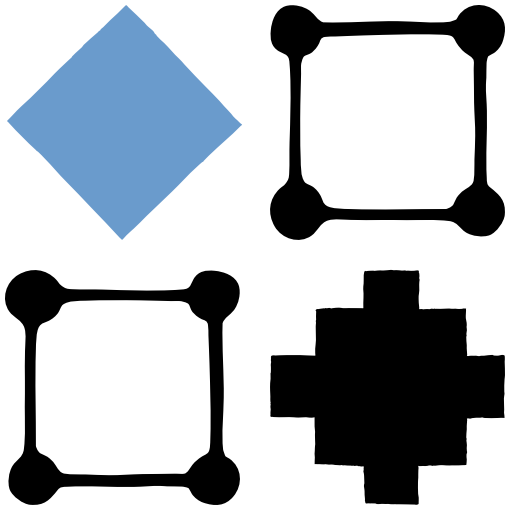

Harness design for long-running application development

Harness design is key to performance at the fron
pushed Claude further in frontend design and lor
engineering.

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Written by Prithvi Rajasekaran, a member of our Labs team.

Over the past several months I've been working on two interconnected problems: getting Claude to produce high-quality frontend designs, and getting it to build complete applications without human intervention. This work originated with earlier efforts on our frontend design skill and long-running coding agent harness, where my colleagues and I were able to improve Claude's performance well above baseline through prompt engineering and harness design—but both eventually hit ceilings.

To break through, I sought out novel AI engineering approaches that held across two quite different domains, one defined by subjective taste, the other by verifiable correctness and usability. Taking inspiration from Generative Adversarial Networks (GANs), I designed a multi-agent structure with a **generator** and **evaluator** agent. Building an evaluator that graded outputs reliably—and with taste—meant first developing a set of criteria that could turn subjective judgments like "is this design good"

I then applied these techniques to long-runnir over two lessons from our earlier harness worl tractable chunks, and using structured artifact sessions. The final result was a three-agent arc evaluator—that produced rich full-stack appli autonomous coding sessions.

Why naive implementations fall short

We've previously shown that harness design has a substantial impact on the effectiveness of long running agentic coding. In an earlier experiment, we used an initializer agent to decompose a product spec into a task list, and a coding agent that implemented the tasks one feature at a time before handing off artifacts to carry context across sessions. The broader developer community has converged on similar insights, with approaches like the "Ralph Wiggum" method using hooks or scripts to keep agents in continuous iteration cycles.

But some problems remained persistent. For more complex tasks, the agent still tends to go off the rails over time. While decomposing this issue, we observed two common failure modes with agents executing these sorts of tasks.

First is that models tend to lose coherence on lengthy tasks as the context window fills (see our post on context engineering). Some models also exhibit "context anxiety," in which they begin wrapping up work prematurely as they approach what they believe is their context limit. Context resets—clearing the context window entirely and starting a fresh agent, combined with a structured handoff that carries the previous agent's state and the next steps—addresses both these issues.

This differs from compaction, where earlier pa summarized in place so the same agent can ke While compaction preserves continuity, it doe which means context anxiety can still persist. the cost of the handoff artifact having enough the work cleanly. In our earlier testing, we fou context anxiety strongly enough that compact enable strong long task performance, so conte

harness design. This solves the core issue, but adds orchestration complexity, token overhead, and latency to each harness run.

A second issue, which we haven't previously addressed, is self-evaluation. When asked to evaluate work they've produced, agents tend to respond by confidently praising the work—even when, to a human observer, the quality is obviously mediocre. This problem is particularly pronounced for subjective tasks like design, where there is no binary check equivalent to a verifiable software test. Whether a layout feels polished or generic is a judgment call, and agents reliably skew positive when grading their own work.

However, even on tasks that do have verifiable outcomes, agents still sometimes exhibit poor judgment that impedes their performance while completing the task. Separating the agent doing the work from the agent judging it proves to be a strong lever to address this issue. The separation doesn't immediately eliminate that leniency on its own; the evaluator is still an LLM that is inclined to be generous towards LLM-generated outputs. But tuning a standalone evaluator to be skeptical turns out to be far more tractable than making a generator critical of its own work, and once that external feedback exists, the generator has something concrete to iterate against.

Frontend design: making subjective q

I started by experimenting on frontend design was most visible. Absent any intervention, Cla safe, predictable layouts that are technically fu unremarkable.

Two insights shaped the harness I built for frontend design. First, while aesthetics can't be fully reduced to a score—and individual tastes will always vary—they can be improved with grading criteria that encode design principles and preferences. "Is this design beautiful?" is hard to answer consistently, but "does this follow our principles for good design?" gives Claude something concrete to grade against. Second, by separating frontend generation from frontend grading, we can create a feedback loop that drives the generator toward stronger outputs.

With this in mind, I wrote four grading criteria that I gave to both the generator and evaluator agents in their prompts:

- **Design quality:** Does the design feel like a coherent whole rather than a collection of parts? Strong work here means the colors, typography, layout, imagery, and other details combine to create a distinct mood and identity.
- **Originality:** Is there evidence of custom decisions, or is this template layouts, library defaults, and AI-generated patterns? A human designer should recognize deliberate creative choices. Unmodified stock components—or telltale signs of AI generation like purple gr
- **Craft:** Technical execution: typography hie harmony, contrast ratios. This is a competer check. Most reasonable implementations d means broken fundamentals.
- **Functionality:** Usability independent of ae what the interface does, find primary actior guessing?

I emphasized design quality and originality ov already scored well on craft and functionality by default, as the required

technical competence tended to come naturally to the model. But on design and originality, Claude often produced outputs that were bland at best. The criteria explicitly penalized highly generic "AI slop" patterns, and by weighting design and originality more heavily it pushed the model toward more aesthetic risk-taking.

I calibrated the evaluator using few-shot examples with detailed score breakdowns. This ensured the evaluator's judgment aligned with my preferences, and reduced score drift across iterations.

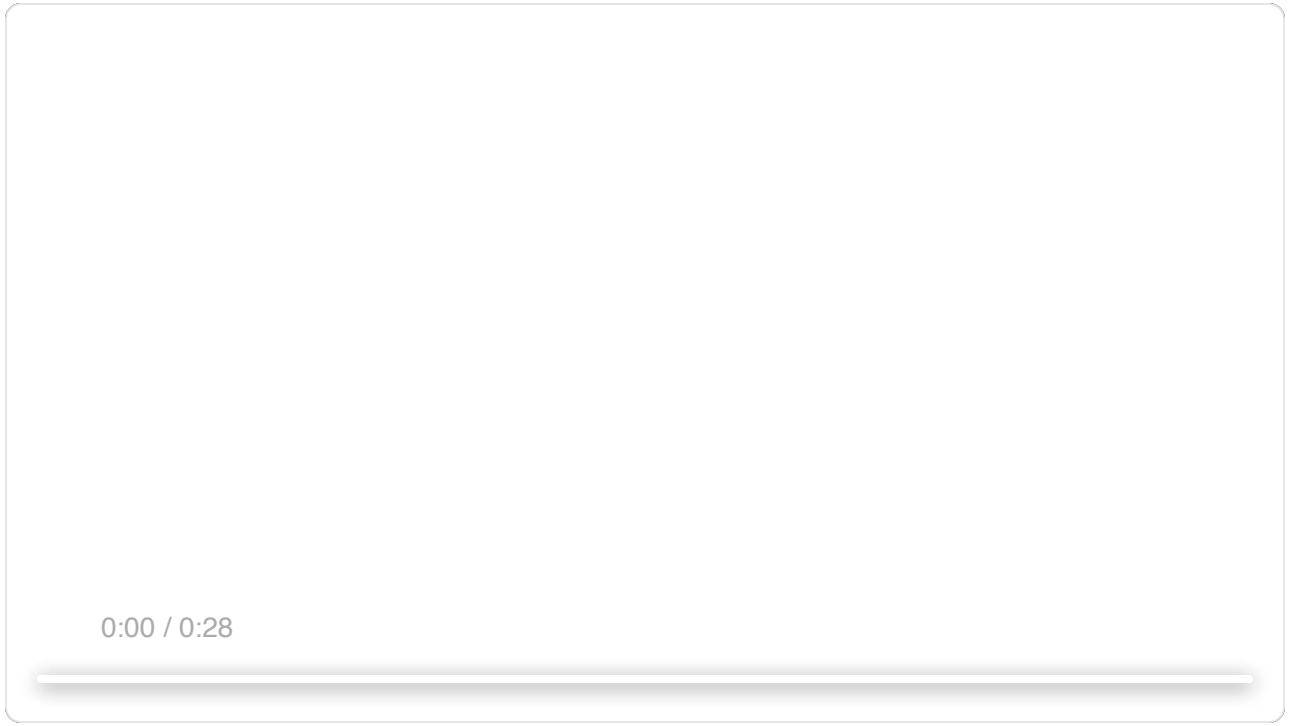
I built the loop on the Claude Agent SDK, which kept the orchestration straightforward. A generator agent first created an HTML/CSS/JS frontend based on a user prompt. I gave the evaluator the Playwright MCP, which let it interact with the live page directly before scoring each criterion and writing a detailed critique. In practice, the evaluator would navigate the page on its own, screenshotting and carefully studying the implementation before producing its assessment. That feedback flowed back to the generator as input for the next iteration. I ran 5 to 15 iterations per generation, with each iteration typically pushing the generator in a more distinctive dir evaluator's critique. Because the evaluator was rather than scoring a static screenshot, each cy runs stretched up to four hours. I also instructe strategic decision after each evaluation: refine trending well, or pivot to an entirely different a working.

Across runs, the evaluator's assessments impr plateauing, with headroom still remaining. So incrementally. Others took sharp aesthetic turns between iterations.

The wording of the criteria steered the generator in ways I didn't fully anticipate. Including phrases like "the best designs are museum quality" pushed designs toward a particular visual convergence, suggesting that the prompting associated with the criteria directly shaped the character of the output.

While scores generally improved over iterations, the pattern was not always cleanly linear. Later implementations tended to be better as a whole, but I regularly saw cases where I preferred a middle iteration over the last one. Implementation complexity also tended to increase across rounds, with the generator reaching for more ambitious solutions in response to the evaluator's feedback. Even on the first iteration, outputs were noticeably better than a baseline with no prompting at all, suggesting the criteria and associated language themselves steered the model away from generic defaults before any evaluator feedback led to further refinement.

In one notable example, I prompted the model to create a website for a Dutch art museum. By the ninth iteration, it had produced a clean, dark-themed landing page for a fictional museum. The page was visu
with my expectations. Then, on the tenth cycl
and reimagined the site as a spatial experienc
rendered in CSS perspective, artwork hung on
and doorway-based navigation between galle
was the kind of creative leap that I hadn't seen
generation.



Scaling to full-stack coding

With these findings in hand, I applied this GA
development. The generator-evaluator loop m
development lifecycle, where code review and
as the design evaluator.

The architecture

In our earlier long-running harness, we had so
coding with an initializer agent, a coding agen
and context resets between sessions. Context resets were a key unlock: the

harness used Sonnet 4.5, which exhibited the "context anxiety" tendency mentioned earlier. Creating a harness that worked well across context resets was key to keeping the model on task. Opus 4.5 largely removed that behavior on its own, so I was able to drop context resets from this harness entirely. The agents were run as one continuous session across the whole build, with the Claude Agent SDK's automatic compaction handling context growth along the way.

For this work I built on the foundation from the original harness with a three-agent system, with each agent addressing a specific gap I'd observed in prior runs. The system contained the following agent personas:

Planner: Our previous long-running harness required the user to provide a detailed spec upfront. I wanted to automate that step, so I created a planner agent that took a simple 1-4 sentence prompt and expanded it into a full product spec. I prompted it to be ambitious about scope and to stay focused on product context and high level technical design rather than detailed technical implementation. This emphasis was due to the concern that if the planner tried to specify granular technical details upfront and got something wr[illegible] cascade into the downstream implementation the agents on the deliverables to be produced they worked. I also asked the planner to find o into the product specs. (See example in the Ap

Generator: The one-feature-at-a-time approac well for scope management. I applied a simila generator to work in sprints, picking up one fe sprint implemented the app with a React, Vite PostgreSQL) stack, and the generator was inst the end of each sprint before handing off to QA. It also had git for version control.

Evaluator: Applications from earlier harnesses often looked impressive but still had real bugs when you actually tried to use them. To catch these, the evaluator used the Playwright MCP to click through the running application the way a user would, testing UI features, API endpoints, and database states. It then graded each sprint against both the bugs it had found and a set of criteria modeled on the frontend experiment, adapted here to cover product depth, functionality, visual design, and code quality. Each criterion had a hard threshold, and if any one fell below it, the sprint failed and the generator got detailed feedback on what went wrong.

Before each sprint, the generator and evaluator negotiated a sprint contract: agreeing on what "done" looked like for that chunk of work before any code was written. This existed because the product spec was intentionally high-level, and I wanted a step to bridge the gap between user stories and testable implementation. The generator proposed what it would build and how success would be verified, and the evaluator reviewed that proposal to make sure the generator was building the right thing. The two iterated until they agreed.

Communication was handled via files: one age would read it and respond either within that fi previous agent would read in turn. The genera upon contract before handing the work off to (the spec without over-specifying implementa

Running the harness

For the first version of this harness, I used Clau prompts against both the full harness and a sii

I used Opus 4.5 since this was our best coding model when I began these experiments.

I wrote the following prompt to generate a retro video game maker:

> *Create a 2D retro game maker with features including a level editor, sprite editor, entity behaviors, and a playable test mode.*

The table below shows the harness type, length it ran for, and the total cost.

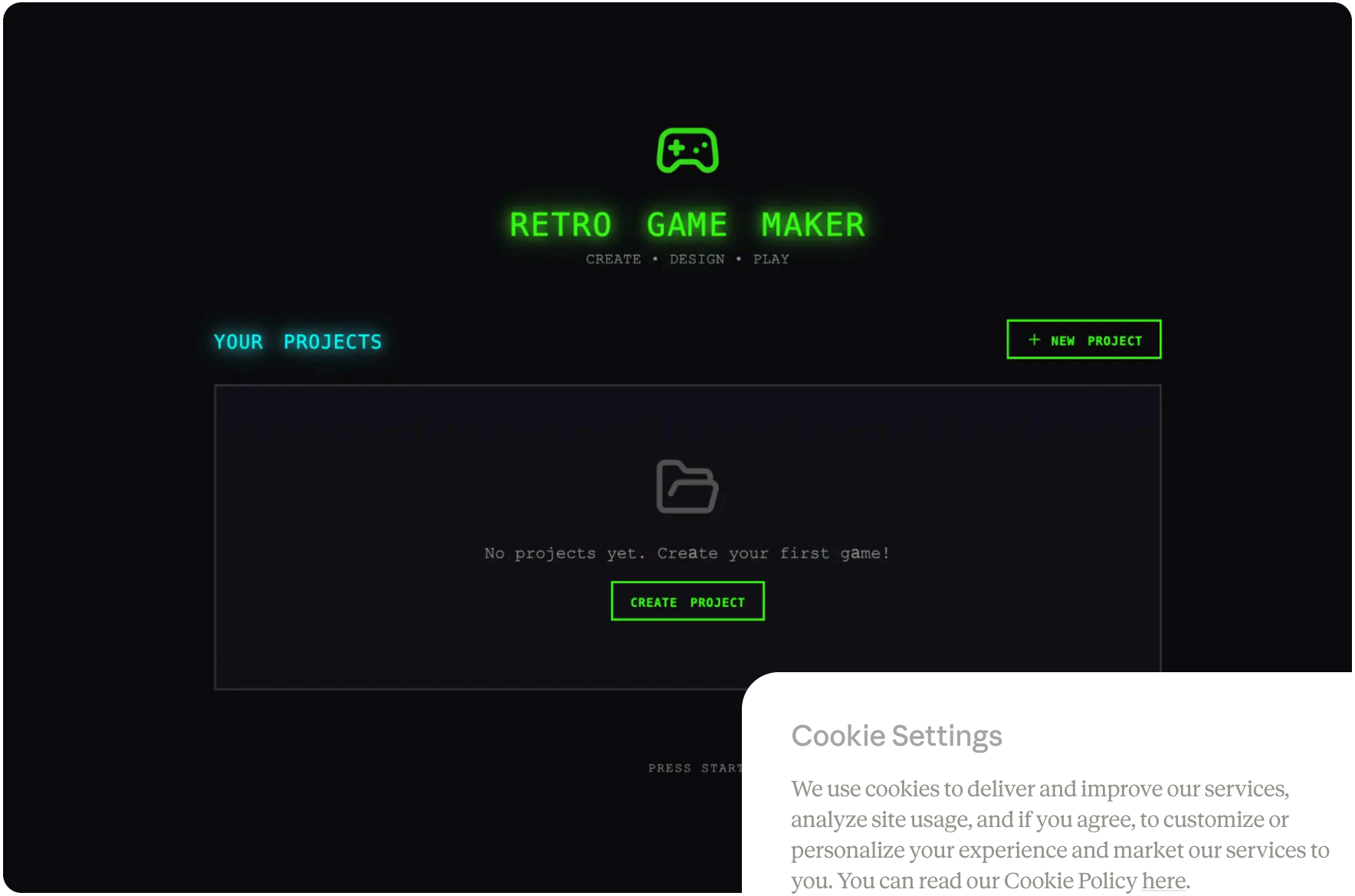
Harness	Duration	Cost
Solo	20 min	$9
Full harness	6 hr	$200

The harness was over 20x more expensive, but the difference in output quality was immediately apparent.

I was expecting an interface where I could construct a level and its component parts (sprites, entities, tile layout) then hit play to actually play the level. I started by opening the solo run's output, and the initial application seemed in line with those expectations.

As I clicked through, however, issues started to emerge. The layout wasted space, with fixed-height panels leaving most of the viewport empty. The workflow was rigid. Trying to populate a level prompted me to create sprites and entities first, but nothing in the UI guided me toward that sequence. More to the point, the actual game was broken. My entities appeared on screen but nothing responded to input. Digging into the code revealed that the wiring between entity definitions and the game runtime was broken, with no surface indication of where.

Opening screen | Sprite editor | Game play



Initial screen when opening the app created by the solo harness.

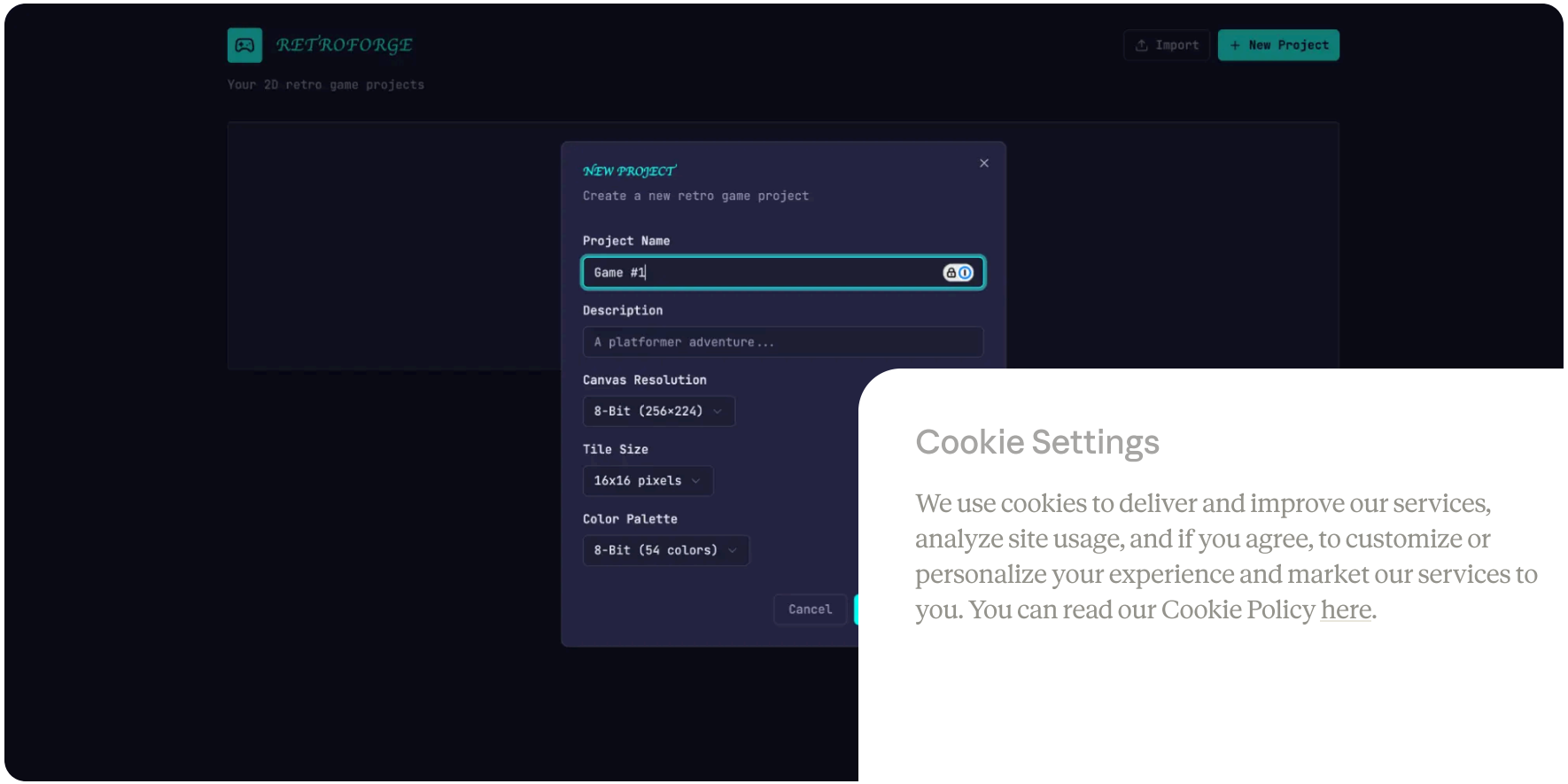
After evaluating the solo run, I turned my attention to the harness run. This run started from the same one-sentence prompt, but the planner step expanded that prompt into a 16-feature spec spread across ten sprints. It went well beyond what the solo run attempted. In addition to the core editors and play mode, the spec called for a sprite animation system, behavior templates, sound effects and music, an AI-assisted sprite generator and level designer, and game export with shareable links. I gave the planner access to our frontend design skill, which it read and used to create a visual design language for the app as part of the spec. For each sprint, the generator and evaluator negotiated a contract defining the specific implementation details for the sprint, and the testable behaviors that would be tested to verify completion.

The app immediately showed more polish and smoothness than the solo run. The canvas used the full viewport, the panels were sized sensibly, and the interface had a consistent visual identity that tracked the design direction from the spec. Some of the clunkiness I'd seen in the
workflow still didn't make it clear that you sho
before trying to populate a level, and I had to fi
This read as a gap in the base model's product
the harness was designed to address, though i
targeted iteration inside the harness could hel
quality.

Working through the editors, the new run's ad
apparent. The sprite editor was richer and mor
palettes, a better color picker, and more usable

Because I'd asked the planner to weave AI features into its specs, the app also came with a built-in Claude integration that let me generate different parts of the game through prompting. This significantly sped up the workflow.

Opening screen | Sprite editor | AI game design | AI game design | Game play



Initial screen: Creating a new game, in the app built with the full harness

The biggest difference was in play mode. I was actually able to move my entity and play the game. The physics had some rough edges—my character jumped onto a platform but ended up overlapping with it, which felt intuitively wrong—but the core thing worked, which the solo run did not manage. After moving around a bit, I did hit some limitations with the There was a large wall that I wasn't able to jum suggested there were some common sense im the harness could handle to further refine the

Reading through the logs, it was clear that the in line with the spec. Each sprint, it walked thr criteria and exercised the running application against anything that diverged from expected granular—Sprint 3 alone had 27 criteria coveri

evaluator's findings were specific enough to act on without extra investigation. The table below shows several examples of issues our evaluator identified:

Contract criterion	Evaluator finding
Rectangle fill tool allows click-drag to fill a rectangular area with selected tile	**FAIL** — Tool only places tiles at drag start/end points instead of filling the region. `fillRectangle` function exists but isn't triggered properly on mouseUp.
User can select and delete placed entity spawn points	**FAIL** — Delete key handler at `LevelEditor.tsx:892` requires both `selection` and `selectedEntityId` to be set, but clicking an entity only sets `selectedEntityId`. Condition should be `selection || (selectedEntityId &&`
User can reorder animation frames via API	**FAIL** — `PUT /frames/reorder` route defined `/{frame_id}` routes. FastAPI matches 'r` eor` integer and returns 422: "unable to parse string a

Getting the evaluator to perform at this level took work. Out of the box, Claude is a poor QA agent. In early runs, I watched it identify legitimate issues, then talk itself into deciding they weren't a big deal and approve the work anyway. It also tended to test superficially, rather than probing edge cases, so more subtle bugs often slipped through. The tuning loop was to read the evaluator's logs, find examples where its judgment diverged from mine, and update the QAs prompt to solve for those issues. It took several rounds of this development loop before the evaluator was grading in a way that I found reasonable. Even then, the harness output showed the limits of the model's QAing capabilities: small layout issues, interactions that felt unintuitive in places, and undiscovered bugs in more deeply nested features that the evaluator hadn't exercised thoroughly. There was clearly more verification headroom to capture with further tuning. But compared to the solo run, where the central feature of the application simply didn't work, the lift was obvious.

Iterating on the harness

The first set of harness results was encouragin
expensive. The logical next step was to find wa
degrading its performance. This was partly co
of a more general principle: every component
assumption about what the model can't do on
worth stress testing, both because they may b
quickly go stale as models improve. Our blog p
frames the underlying idea as "find the simple
increase complexity when needed," and it's a p
for anyone maintaining an agent harness.

In my first attempt to simplify, I cut the harness back radically and tried a few creative new ideas, but I wasn't able to replicate the performance of the original. It also became difficult to tell which pieces of the harness design were actually load-bearing, and in what ways. Based on that experience, I moved to a more methodical approach, removing one component at a time and reviewing what impact it had on the final result.

As I was going through these iteration cycles, we also released Opus 4.6, which provided further motivation to reduce harness complexity. There was good reason to expect 4.6 would need less scaffolding than 4.5 did. From our launch blog: "[Opus 4.6] plans more carefully, sustains agentic tasks for longer, can operate more reliably in larger codebases, and has better code review and debugging skills to catch its own mistakes." It also improved substantially on long-context retrieval. These were all capabilities the harness had been built to supplement.

Removing the sprint construct

I started by removing the sprint construct enti
helped to decompose work into chunks for the
the improvements in Opus 4.6, there was gooc
could natively handle the job without this sort

I kept both the planner and evaluator, as each
Without the planner, the generator under-scop
start building without first speccing its work, a
rich application than the planner did.

With the sprint construct removed, I moved th
end of the run rather than grading per sprint. Since the model was much more

capable, it changed how load-bearing the evaluator was for certain runs, with its usefulness depending on where the task sat relative to what the model could do reliably on its own. On 4.5, that boundary was close: our builds were at the edge of what the generator could do well solo, and the evaluator caught meaningful issues across the build. On 4.6, the model's raw capability increased, so the boundary moved outward. Tasks that used to need the evaluator's check to be implemented coherently were now often within what the generator handled well on its own, and for tasks within that boundary, the evaluator became unnecessary overhead. But for the parts of the build that were still at the edge of the generator's capabilities, the evaluator continued to give real lift.

The practical implication is that the evaluator is not a fixed yes-or-no decision. It is worth the cost when the task sits beyond what the current model does reliably solo.

Alongside the structural simplification, I also added prompting to improve how the harness built AI features into each app, specifically getting the generator to build a proper agent that could drive the app's
That took real iteration, since the relevant kno
Claude's training data covers it thinly. But witl
building agents correctly.

Results from the updated harness

To put the updated harness to the test, I used t
Digital Audio Workstation (DAW), a music pro
recording, and mixing songs:

> *Build a fully featured DAW in the browser usi*

The run was still lengthy and expensive, at about 4 hours and $124 in token costs.

Most of the time went to the builder, which ran coherently for over two hours without the sprint decomposition that Opus 4.5 had needed.

Agent & Phase	Duration	Cost
Planner	4.7 min	$0.46
Build (Round 1)	2 hr 7 min	$71.08
QA (Round 1)	8.8 min	$3.24
Build (Round 2)	1 hr 2 min	$36.89

QA (Round 2)
6.8 min
$3.09

Build (Round 3)
10.9 min
$5.88

QA (Round 3)
9.6 min
$4.06

Total V2 Harness
3 hr 50 min
$124.70

As with the previous harness, the planner exp
full spec. From the logs, I could see the genera
the app and the agent design, wiring the agent
off to QA.

That being said, the QA agent still caught real
noted:

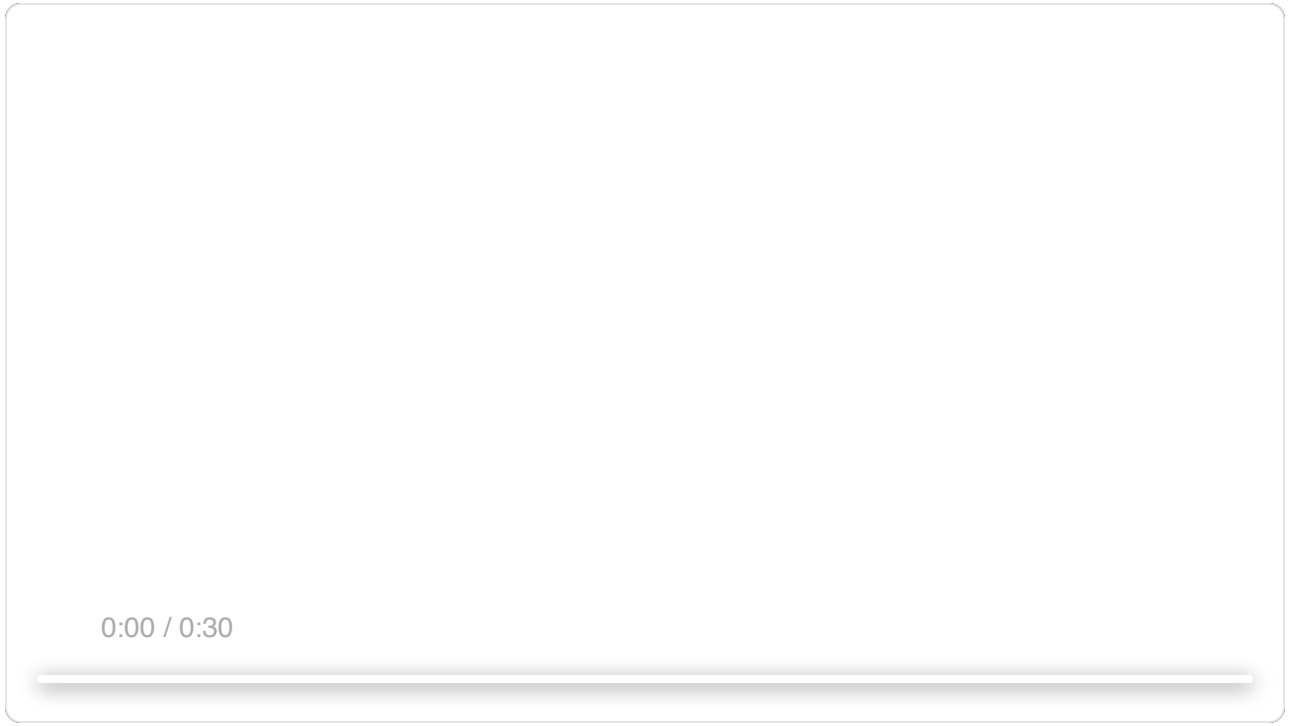
> This is a strong app with excellent design fidelity, solid AI agent, and good backend. The main failure point is Feature Completeness — while the app looks impressive and the AI integration works well, several core DAW features are display-only without interactive depth: clips can't be dragged/moved on the timeline, there are no instrument UI panels (synth knobs, drum pads), and no visual effect editors (EQ curves, compressor meters). These aren't edge cases — they're the core interactions that make a DAW usable, and the spec explicitly calls for them.

In its second round feedback, it again caught several functionality gaps:

> Remaining gaps:
> - Audio recording is still stub-only (button toggles but no mic capture)
> - Clip resize by edge drag and clip split not implemented
> - Effect visualizations are numeric sliders, not graphical (no EQ curve)

The generator was still liable to miss details or stub features when left to its own devices, and the QA still added value in catchir
generator to fix.

Based on the prompt, I was expecting a progra
harmonies, and drum patterns, arrange them i
integrated agent along the way. The video belc



The app is far from a professional music produ
song composition skills could clearly use a lot
actually hear, which made the QA feedback lo
musical taste.

But the final app had all the core pieces of a fu
program: a working arrangement view, mixer,
browser. Beyond that, I was able to put togethe
through prompting: the agent set the tempo a
drum track, adjusted mixer levels, and added r
composition were present, and the agent coul

drive them autonomously, using

tools to create a simple production from end to end. You might say it's not pitch-perfect yet—but it's getting there.

What comes next

As models continue to improve, we can roughly expect them to be capable of working for longer, and on more complex tasks. In some cases, that will mean the scaffold surrounding the model matters less over time, and developers can wait for the next model and see certain problems solve themselves. On the other hand, the better the models get, the more space there is to develop harnesses that can achieve complex tasks beyond what the model can do at baseline.

With this in mind, there are a few lessons from this work worth carrying forward. It is always good practice to experiment with the model you're building against, read its traces on realistic problems, and tune its performance to achieve your desired outcomes. When working on more complex tasks, there is sometimes headroom from decomposing the task and applying specialized agents to each aspect of the problem. And when a new model lands, it is generally good practice to re-examine a harness, stripping away pieces that are no longer load-bearing to performance and adding new pieces to achieve greater capability that may not have been possible before.

From this work, my conviction is that the space of interesting harness combinations doesn't shrink as models improve. Instead, it moves, and the most interesting work for AI engineers is to keep finding the next novel combination.

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Appendix

Example plan generated by planner agent.

```
RetroForge - 2D Retro Game Maker

Overview
RetroForge is a web-based creative stu
building 2D retro-style video games. I
charm of classic 8-bit and 16-bit game
intuitive editing tools—enabling anyor
indie developers to bring their game i
traditional code.

The platform provides four integrated
based Level Editor for designing game
Editor for crafting visual assets, a v
for defining game logic, and an instan
```

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